



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SELVI AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 120.41 km
Estimated Duration: 2.3 hours
Adjusted Duration (Heavy Vehicle): 2.8 hours
Start: (11.4381, 77.8734)
End: (11.5248, 78.7572)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

The route from Sangagiri, Tamil Nadu to Navalur, Tamil Nadu, covers approximately 120.41 kilometers. It primarily follows state highways and local roads, connecting rural and semi-urban areas. The journey usually takes around 2.26 hours for heavy vehicles. The drive encompasses a mix of flat terrains and minor elevation changes, with agricultural lands and small towns.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather:** Tamil Nadu typically experiences a tropical climate with high temperatures and humidity. The southwest (June to September) and northeast (October to December) monsoons bring heavy rainfall that may cause flooding and road blockage.
- **Hazards:** During monsoon seasons, the risk of slippery roads and reduced visibility is high. Waterlogging and landslides on certain stretches can occur, particularly in lower-lying areas.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic congestion is common during morning (7-9 AM) and evening (5-7 PM) in towns like Salem and smaller junctions.
- **Congestion Areas:** Bottlenecks typically occur at intersections linking major highways and local roads, especially near Salem, and small towns en route, due to mixed traffic including public transport and commercial vehicles.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Varies significantly along the route; highways generally have good-quality pavement, but local roads may suffer from potholes and uneven surfaces.
- **Infrastructure:** Adequate signage and lighting are present on highways, but may be deficient on rural roads. Some stretches may lack hard shoulders and adequate barricades.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies or obstructions, alternative routing through NH544 and NH79 may provide detours around congested or unsafe segments, although a longer distance may be involved.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials requires adherence to specific permits and safety guidelines as per Tamil Nadu and central Indian laws.
- Use of designated routes, timing restrictions, and safety bottlenecks must be followed to minimize risk.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Previous incidents in the region often relate to over-speeding and vehicular malfunction. Although rare, hazardous material spillages near farmland can have severe environmental consequences.

8. Environmental Considerations and Sensitive Areas

- Important zones include agricultural lands and small water bodies, which can be affected by spillage or pollution.
- Areas near large green cover or water catchment areas should be approached cautiously to prevent ecological disruption.

9. Analysis of Communication Coverage

- Mobile coverage is generally reliable on highways and near towns. However, rural stretches may experience short dead zones affecting communication.
- Drivers are advised to be prepared with offline navigation and emergency contact information.

10. Estimated Emergency Response Times

- Urban areas like Salem typically have faster emergency response times (15-30 minutes).
- Remote areas may experience delayed responses due to distance and lesser infrastructure support (up to 1 hour or more).

11. Overall Summary of Risk Assessment

The route features a mix of urban and rural challenges. Monsoon weather poses significant risks, particularly relating to road conditions and visibility. While most of the route is covered by reliable communication networks, certain zones may face connectivity issues, affecting timely support in emergencies. Compliance with transport regulations for hazardous materials is crucial to preventing incidents. Overall, while navigable, this route requires careful planning and alertness, especially concerning road quality and weather conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.53501, 77.95707	30 KM/Hr
13	Turn	Medium	11.53601, 77.95856	30 KM/Hr
14	Turn	Medium	11.64172, 78.11817	30 KM/Hr
15	Turn	Medium	11.64216, 78.11875	30 KM/Hr
16	Turn	High	11.64195, 78.12129	15 KM/Hr
17	Turn	High	11.64120, 78.12168	15 KM/Hr
18	Turn	Medium	11.63948, 78.12081	30 KM/Hr
19	Turn	Medium	11.65703, 78.20465	30 KM/Hr
20	Turn	High	11.58562, 78.74231	15 KM/Hr
21	Turn	High	11.58528, 78.74234	15 KM/Hr
22	Turn	High	11.58553, 78.74952	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
22	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
23	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
24	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
25	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
26	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
27	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
28	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
29	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
30	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
31	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
35	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
37	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
38	hospital	Amsb Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium
39	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium
40	hospital	Government Hospital, Kothampadi	11.62641, 78.542199	30 km/h	Medium
45	clinic	Sugam Clinic	11.5832481, 78.7496438	30 km/h	Medium
46	hospital	Selvam Hospital	11.585978, 78.748761	30 km/h	Medium
47	hospital	Prabushankar Hospital	11.584239, 78.7490323	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
21	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
32	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
33	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
34	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium
41	school	Vani Vikas Matric Hr.sec school	11.6212713, 78.5462658	30 km/h	Medium
42	school	Saraswati Metric Hr sec School	11.5910827, 78.6402264	30 km/h	Medium
43	college	Maruthi Polytechnic College	11.5856852, 78.7201072	30 km/h	Medium
44	college	Bharathiyar Educational Institure	11.5863463, 78.7364793	30 km/h	Medium

Route Photos of Risky Spots



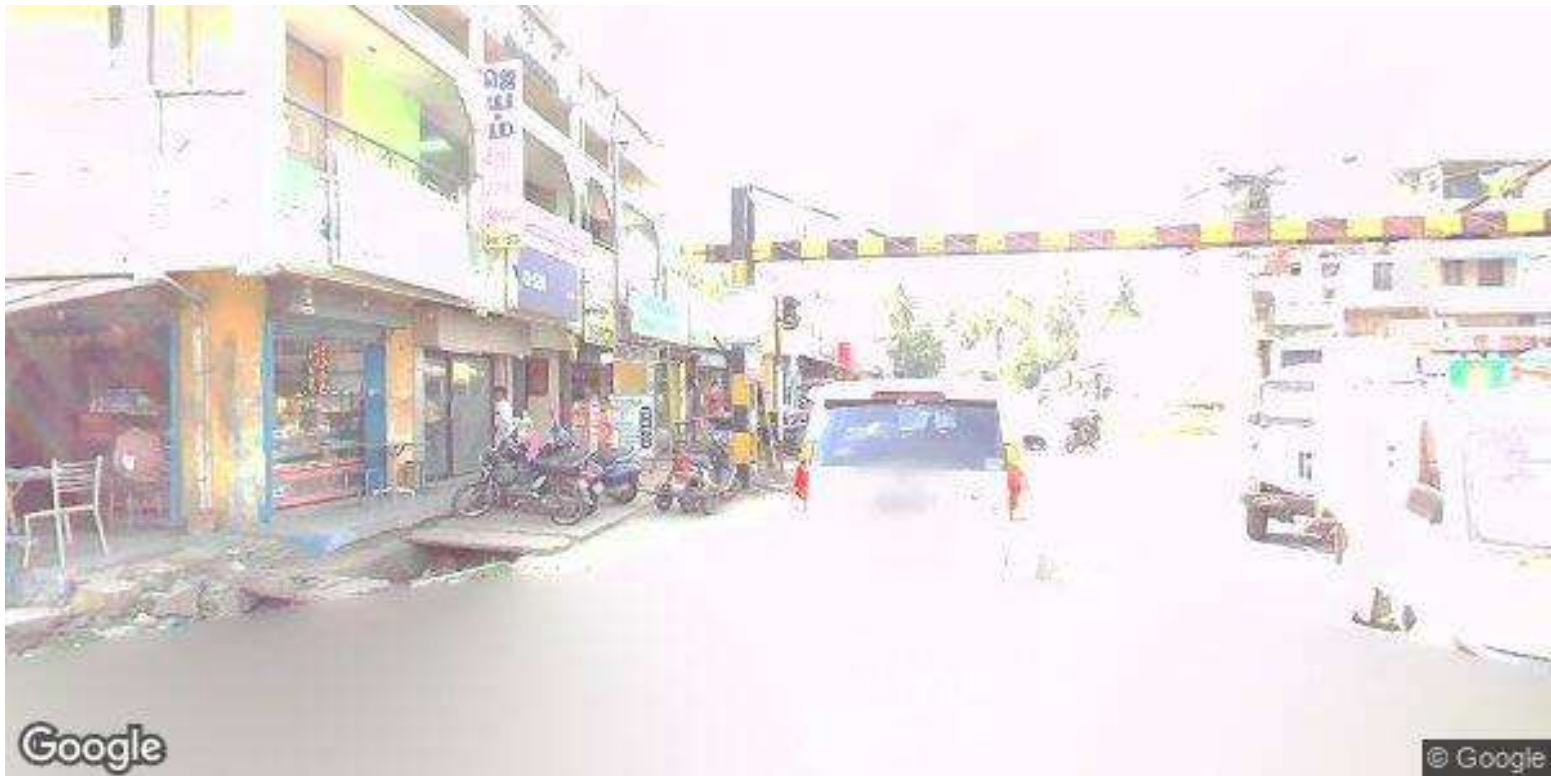
Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64216, 78.11875



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64195, 78.12129



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65703, 78.20465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58562, 78.74231



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58528, 78.74234



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58553, 78.74952

